

Numbers

Writing Numbers

1. Words versus numerals

1.1 Use words for numbers one through nine and numerals for larger numbers:

four horses

seven ships

but: 14 people, 10 oranges

1.2 Use words for expressions used in a mathematical sense if the quantity discussed is less than 10:

fivefold

by three orders of magnitude

five times as many as

but: 12 times as many

10-fold

Usage: See last section of this document. Stated succinctly, the term "times faster than" or "times more than" could be taken to mean a multiplicative increase ($A = 500B$) or an additive increase ($A = 500B + B$).

For "times more than" versus "times as much as," for example, 100 times as much as (100 times x) rather than 100 times more than, because the latter means an increase 100 times in addition to (100 times $x + x$). To illustrate this, you can imagine a one-story, a four-story (three times more), and a three-story (three times as high as) building.

"Times less than" does not make sense mathematically; ask the author for a percentage. See last section.

1.3 Use numerals to express a number used with a standard unit of measure. Insert a space between the numeral and the unit, except for %, ° (degrees), ' (angular minutes), and '' (angular seconds).

10 km	25 mg/kg	25%
24 hours	220°C	30°11'22''

1.4 In a series or range containing some numbers greater than 9 and some numbers less than 10, use numerals for all.

Each of the five animals received 7 injections of substance A and 14 injections of substance B.

("Five" is spelled out because it is not part of the series and "animals" is not a unit. We would use a numeral for more than nine animals.)

2. Multiple-digit numbers

- 2.1 In numbers consisting of two to four digits and in dates, run the numerals together. Follow this rule even when a four-digit number appears in a series with numbers containing five or more digits (see section 2.3 for how to handle in tables).

4280, 1492
2349, 10,000, and 57,123

- 2.2 Numbers consisting of five or more numerals contain commas:

10,000, 238,527

However, we do not insert commas in five-digit page numbers in references.

- 2.3 When one number in a column of numbers in a table contains five or more numerals, put commas in those containing four also.

57,123
2,349
10,000

- 2.4 Dates: 6 August 1946

- 2.5 Time: 1700 or 9:00 a.m.

- 2.6 Page range: pp. 33–69

- 2.7 Percentage: 99.773%, 6%, 8 to 10%

- 2.8 Decimal quantity: 6.67, 0.01 (always use an initial zero for numbers less than 1)

- 2.9 Numerical designation: magnification, $\times 50$
a five-point scale from 0 to 4
a factor of 4
18 multiplied by 2

3. Ordinal numbers

- 3.1 Treat ordinal numbers as you would cardinal numbers. Mark the “th,” “rd,” etc., so that they are on the line, not superscripts. See the dictionary under numbers.

seventh 23rd 12th one-ninth

second-order reaction 11th-order fitter

3.2 Follow author on use of ordinal numbers in street addresses:

New York University Medical School, 1800 5th Avenue, New York, NY
10032, USA

4. Large numbers

When possible, simplify large numbers containing many initial or terminal zeros by either

- substituting a word:

change \$ 1,015,000,000,000
to \$ 1.015 trillion
or \$1015 billion

- or using an appropriate prefix of power of 10:

change 0.000023 g
to 23 μg
or 2.3×10^{-5} g

(There may be times where one unit may be the “natural” one and a change may reduce readability.)

5. Millions, billions, etc.

Use \$1 million to \$2 million, not \$1 to \$2 million, even if you think it is clear from the context.

Similarly, 5 million to 6 million people

Fractions

1. Avoid using fractions without whole numbers appended to them ($\frac{1}{2}$, $\frac{3}{4}$, etc.) in the text of a paper. If a fraction without whole numbers must be used (as a subscript or exponent, for example), the fraction should be “built.” “Building” a fraction means that the numbers are set on top and to the side of each other ($\frac{1}{2}$), not as $1/2$ or $\frac{1}{2}$.

Example: $a_{\frac{1}{2}}$ (subscript)
 $20^{\frac{1}{2}}$ (exponent)

2. Do not convert a fraction ($\frac{1}{4}$ or one-quarter) to a percent (25%) or a decimal (0.25) unless the original use of the fraction specified by the author is awkward in its context or does not allow an expression to conform to *Science* style.

Awkward: After five and one-half min

Better: After 5.5 min. . . .

or: After $5\frac{1}{2}$ min

3. Spell out and hyphenate fractions whose terms are both less than 10; use a built fraction if one of the terms is 10 or greater:

one-quarter of the population

$\frac{1}{16}$ of the volume

two-thirds of the samples

$\frac{1}{20}$ of the pressure

But:

It is not necessary to make a distinction between “a fifth” and “one-fifth”:

With up to a fifth or more of the species of all groups. . . .

Units of Measurement (based in part on *ACS*)

1. Where possible, use metric and SI units.

Exceptions: - a 10-gallon tank

- certain wire, hypodermic needle, screw, and shotgun
gauges: 15 psi (pounds per square inch)

Occasionally miles, tons, and inches will be allowed (if units based on the English system are used, spell them out).

Sometimes it is better to provide a conversion factor and leave the original unit, as when some measuring device was actually calibrated in non-SI units.

Example: lamberts [$1 \text{ lambert} = 3.183 \times 10^3 \text{ cd/m (candela/meter)}$]

Some non-SI units continue to be used and do not require conversion factors.

Å

hectare

Conversion factors are recommended for the following non-SI unit:

atm ($1 \text{ atm} = 101.325 \text{ kPa}$)

2. Abbreviate the unit of measure when it appears with a numeral and leave a space between the numeral and the unit (this differs from the British style). Hyphenate numbers and units when used as modifiers. Use the word “of” between the unit and the item being measured (exception: molar and osmolar; see below):

75 mg of penicillin	a 75-mg dose of penicillin
0.05 g of NaCl	a 0.05-g sample of NaCl
100 ml of solution	the 100-ml flask

Exception: Difference between molar (M) and mole (mol):

Mole refers directly to an amount; specifically, 6.023×10^{23} molecules of some compound is 1 mole. Therefore, it is correct to say “12 mol of sugar” or “0.30 mol of NaCl” because mole specifies the amount of the compound.

On the other hand, molarity is a unit of concentration based on the amount of some compound (in moles) per volume of something else (in liters). A 2 M HCl solution describes the solution and not the amount of solute in it. Thus, we do not hyphenate or use “of” for a unit signifying molarity (M), osmolarity (osM), molality (*m*), or osmolality (osm) because each of these deals with a concentration within the unit.

0.1 M NaOH	A 0.05 μ M CaCl ₂ solution was added.
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Do not use plurals with abbreviated units:

75 mg, not 75 mgs
2 mol, not 2 mols

3. Compound units

- 3.1 Both the solidus (meaning “per”) and negative exponents are permissible ways of indicating compound units.

Do not use a solidus or negative exponent at all in the item if the denominator is not a standard unit of measure

OK: 4000 cells/ml
4000 g liter⁻¹

Not OK: 4000 cells/well
Change to: 4000 cells per well

Avoid combining the solidus and exponent style within the same paper.

Either style should be used consistently in a unit designation.

Correct: 0.40 g/liter or 0.40 g liter⁻¹

Incorrect: 0.40 g/liter⁻¹

3.2. Some units with negative exponents have no positive ones

“a neutron flux of 500 cm⁻³”

(means 500 neutrons per cubic centimeter of volume)

3.3 Use negative exponent notation when referring to more than two units to avoid ambiguity (do not use multiple soliduses)

change	20 mg/liter/day
to	20 mg liter ⁻¹ day ⁻¹ or 20 mg/liter per day

change	0.1 g/cm/s
to	0.1 g cm ⁻¹ s ⁻¹ or 0.1 g/cm per second

Composite units, for example, 75 ohm·cm and 0.9 A·hour, are written with a center dot between the units. These units may be joined with other units, as:

75 ohm·cm min⁻¹
75 V (ohm·cm)⁻¹

3.4. Sometimes parts of compound units must be spelled out for clarity

Example: 0.2 g/kg of streptomycin

This phrase “0.2 g/kg of streptomycin” could be taken to mean 0.2 g of something per kilogram of streptomycin, which is not the sense usually meant. Rather, this phrase means “0.2 g of streptomycin per kilogram of body weight.” Thus, we would spell out the compound unit the first time in this manner; thereafter, we would use “streptomycin (0.2 g/kg).”

Another example: 5×10^6 hydrogen atoms per square centimeter per second (first time)

and then: 5×10^6 atom cm⁻² s⁻¹

4. Ranges and series

In ranges and series, retain only the final unit of measure, except for degrees and dollars.

at 9:00, 9:15, and 9:30 a.m.
between 13 and 19 years of age
6- and 10-km races
doses of 100, 250, and 500 mg
 $10 \pm 0.002\%$
20- to 30-fold
\$1 million to \$2 million
from 10°N to 20°N
from 2×10^5 to 5×10^5 cells (repeat multiplier)

Degrees, whether of angle or of temperature, are retained for each number.

latitude $22.27^{\circ} \pm 0.002^{\circ}\text{N}$; from 10°S to 10°W
between 10° and 12° of visual angle
temperature ranged from 35° to 40°C

No degree sign is used with kelvin: 250 K

Do not use the Fahrenheit scale.

5. Spelling out numerals and units

When a sentence starts with a specific quantity, spell out the number as well as the unit of measure (*ACS*, ed. 2, p. 144).

Ten milliliters of distilled water was added.

However, if possible, recast the sentence:

Distilled water (10 ml) was added.

The verb “was” is used because units of measurement are treated as collective nouns and therefore take a singular verb.

But

Five to 15 ml were added.

Twenty to 25 ml were added.

6. Concentration

10^{-8} M 0.4 μM 0.1 *m* (molal)

7. Dimensions

Specify the unit for each dimension: a cage 4 cm by 6 cm by 7 cm (do not use the times sign because it designates a mathematical operation).

8. Approximate terms

The vernacular “about” or “around” should not be conflated with the mathematic term “approximately”; do not replace either of the former with the latter, or vice versa, even when it appears clear that the mathematic term is (or isn’t) meant. Best to leave the language alone.

However, when “approximately” precedes a value of measurement, replace it with a tilde “~”: “approximately 10 mm” to “~10 mm” (but leave “about 10 mm” as is).

Do not use “circa” or “ca.”; replace with a tilde “~” in most cases. When referring to approximate dates, then spelling out “approximately” may be preferable:

“ca. 45,000 years” to “~45,000 years”

“from approximately 1100 to 1850 BCE”

Reporting Time

Under the 24-hour system, the day begins at midnight, 0000, and the last minute of the day is 2359. A colon is not necessary (although it may be used) in time designations other than a.m. or p.m.

17:00

9:00 a.m.

15th century

21:00 UT (universal time; dft)

fiscal year (FY) 1993 for first usage; FY 1993 thereafter

2100 GMT (Greenwich Mean Time)

Plurals

Form by adding s.

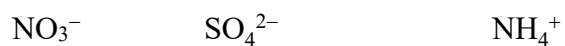
twos and threes

1950s

’50s (most often in News stories)

Subscripts and Superscripts

In chemical formulas, subscripts and superscripts are offset (staggered)



but in mathematical formulas usually are stacked (follow author).

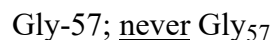
$$x_i^3, \text{ but } T_m^3$$

Amino Acids and Peptides

1. Position of residue: superscript



rather than



2. Fragments of a peptide: set the fragment range in parentheses, and use an en dash; close up the space between the peptide name and opening parenthesis



Positions of atoms

Be sure not to confuse C-3 or C3 (third C in a chain) with C₃ (propane derivative). C₁₈ means that there are 18 carbons in the compound.

Nuclides and Isotopes

The ionic charge, number of atoms, atomic number, and mass number (atomic weight) characterizing the nuclide of an element are indicated by four indices placed around the symbol. Thus,



represents, in clockwise order, a molecule with two positive charges containing two atoms of sulfur, each having atomic number 16 and mass number 32.

A nuclide is indicated by its mass number placed as a superscript to the left of the symbol. For example:



When spelled out, it should correspond to the spoken word (phosphorus-32).

In an isotopically labeled compound, the isotopic prefix in brackets precedes and is attached directly to the part of the name to which it refers:

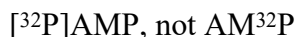
sodium [^{14}C]formate; iodo[1,2- ^{14}C]acetic acid; fructose 1, 6-[1- ^{32}P]bisphosphate
(1 and 2 indicate which carbon or phosphorus atoms are labeled.)

Terms such as “ ^{131}I -labeled albumin” should not be contracted to “[^{131}I]albumin” because native albumin does not contain iodine.

However, when an element is actually part of a molecule, we should always use brackets. That is, [^3H]PYY, [γ - ^{32}P]AMP (the latter is the proper way to designate this compound and others like it; that is, [γ - ^{31}P]ATP, etc.). The Greek letter γ indicates which carbon atom is labeled. One can usually guess whether an element is actually part of the molecule in question, knowing that C, N, and H are almost always part of a biological molecule, whereas I is almost never part of a such molecule.

When an element is only attached to a molecule but is not a part of it, we should use “labeled” the first time and then a hyphen for subsequent references. That is, ^{125}I -labeled fPY (first time), ^{125}I -fPY (subsequent times).

The superscript mass number of an isotope must not be attached to a letter in an abbreviation but precedes the entire abbreviation:



When a substance is labeled by isotopes of more than one element, the symbols are arranged in alphabetical order (including ^2H and ^3H for deuterium and tritium, respectively). When more than one position in a substance is labeled by means of the same isotope, the number of labeled atoms is added as a subscript to the right, as in [$^{14}\text{C}_2$]glycolic acid. If the position of the labeled element is known, it is indicated by an arabic numeral or by an appropriate italic prefix:

[1- ^{14}C]acetic acid; [3- ^{14}C ;2,3- ^2H ; ^{13}N]serine; [*carboxy*- ^{14}C]leucine;
L-[*methyl*- ^{14}C]methionine

If the position of labeling is unknown, the indication is omitted; for example:

[¹⁴C]glucose

If labeling is uniform or general, the indication is

[U-¹⁴C]glucose or [G-¹⁴C]glucose, respectively.

For nuclear reactions, the following form is used (from *ACS*, 1965 edition):

²⁷Al(p,d)²⁶Si

¹⁴N(d,p)¹⁷O

¹²C(n,pγ)¹²N

¹⁰Be(t,n)¹²Be

Astronomy

Use parentheses around the numerical portion of asteroid names

e.g., “(486958) Arrokoth” and “(162173) Ryugu”

Astronomical coordinates

The most common coordinate system is the right ascension and declination (abbreviated “RA” and “dec,” respectively). The right ascension has superscripts indicating hours (“h”), minutes (“m”), and seconds (“s”); the declination has the degree symbol and then prime (for arc minutes) and double prime (for arc seconds). The second and arc second symbols should appear above (ideally directly above) the decimal points. It’s not possible to perfectly stack these symbols in Word, although one can come close using MathType (baseline period followed by a superscript “s” or double prime):

09^h44^m29.^s841, −45°06′35.[″]43 (J2000 equinox)

The alternative is to ask Strive to stack these symbols at the galley correction stage.

The declination always has either a plus or minus sign, RA is always positive, and leading zeros are used. The maximum values for each part are 23:59:59.999... and ±89:59:59.99.... It is common, but not required, for the RA to have one more significant figure than the dec. The equinox must be stated somewhere (usually in the caption or surrounding text)—the most common is J2000, but B1950 and ICRF are also sometimes used.

In any case, colons (e.g., 12:47:56.62 +09:45:05.0) should not be used. The colon notation is common in informal writing but should not appear in published material.

Other astronomical coordinate systems (such as Galactic longitude and latitude, telescope altitude and azimuth, or decimal RA and dec) are expressed solely in degrees and thus do not require the above-noted format.

Arithmetical Comparisons

The use of a comparative muddies the mathematical relation.

Change: seven times larger than
to: seven times as large as

Change: 13 times larger than
to: 13 times as large as

Change: four times fewer than
to: one-fourth as much as or one-fourth

Change: three times higher than predicted
to: three times the predicted number

Correct: there is a 10-fold difference between offshore and onshore concentrations

Correct: there was less than a 10% difference between the. . .

Comment: There are no natural units for weakness, precision, and similar quantities; thus, it is not clear what is meant by “half as weak” or “twice as precise.”

Change: They have a fivefold weaker action at kainate receptors.
to: They are one-fifth as effective at kainate receptors.
or to: Their action at kainate receptors is one-fifth as strong.

Keep in mind that “-fold” means “times as much.” Although the following form is often seen in biological science papers

The outer heavily sclerotized portion of the cuticle contains twofold more protein, 18-fold more chitin, and ninefold less water than unsclerotized cuticle does.

it is better to change this to

The outer heavily sclerotized portion of the cuticle contains twice as much protein, 18 times as much chitin, and one-ninth as much water as unsclerotized cuticle does.